

Enterprise Network Defense: FortiGate Implementation

A Zero-Trust & Next-Gen Firewall Architecture Project

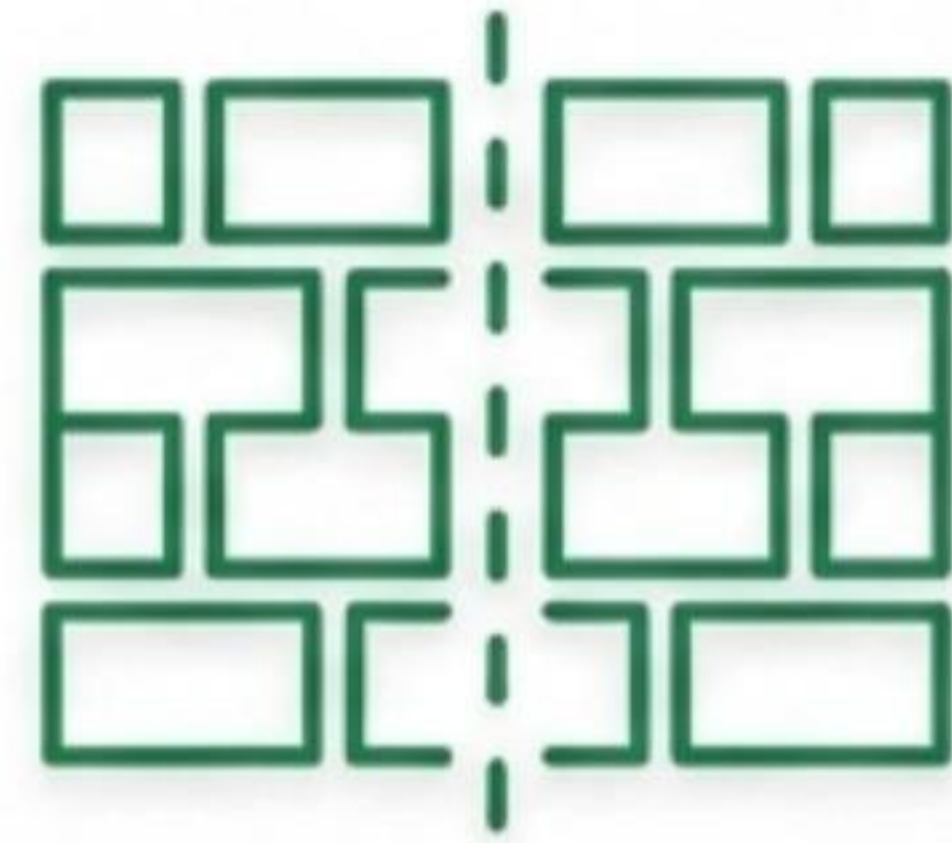
FıratCan Bekar | Network Security Architect Portfolio

Project Architecture & Competencies



Secure Access

SSL-VPN Tunnels
& User
Authentication



Segmentation

Zero-Trust
Isolation for
Linux (SSH/ICMP)
& Windows (RDP)



Traffic Control

Virtual IPs, Port
Forwarding & Layer
7 Application
Filtering

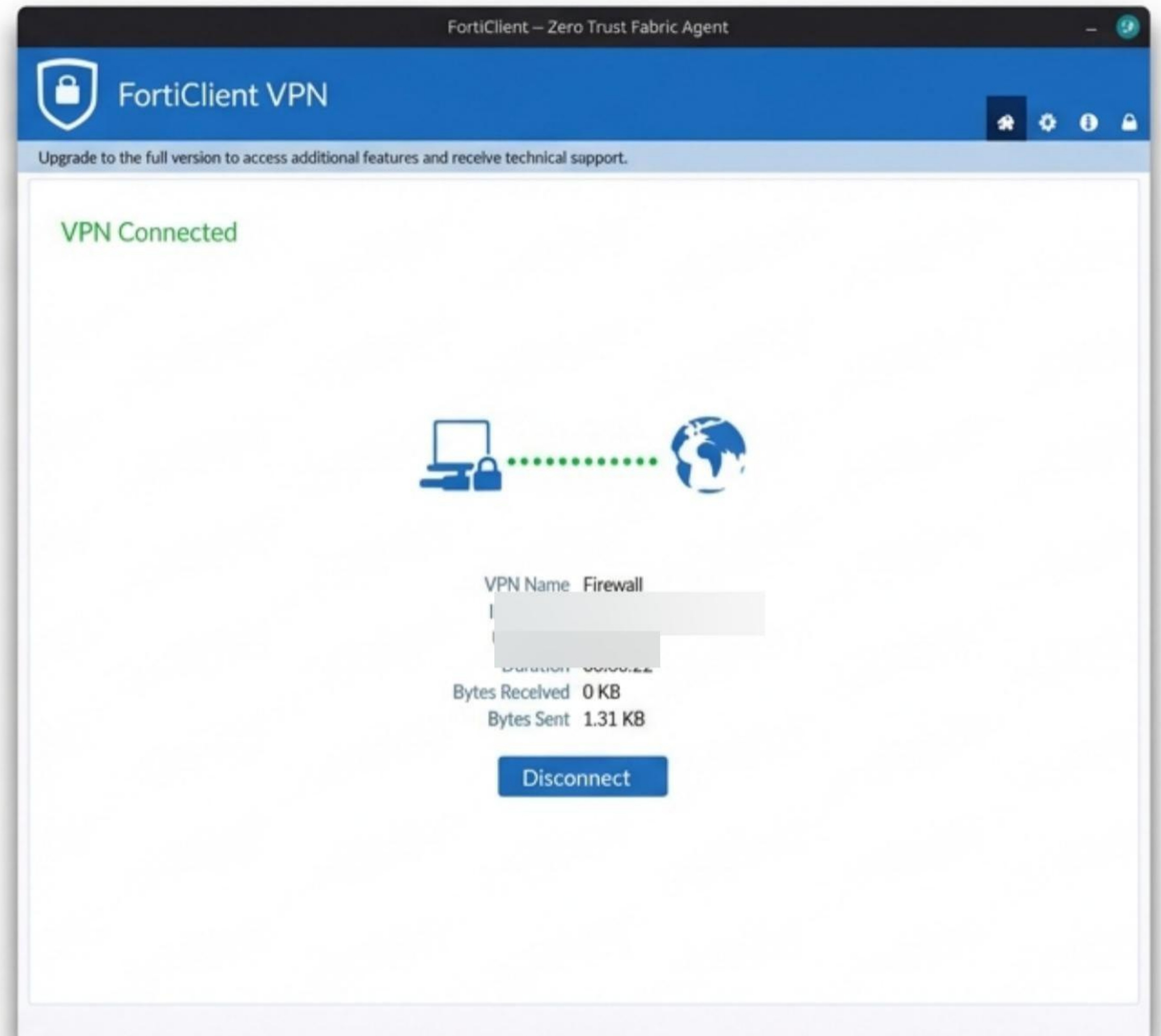
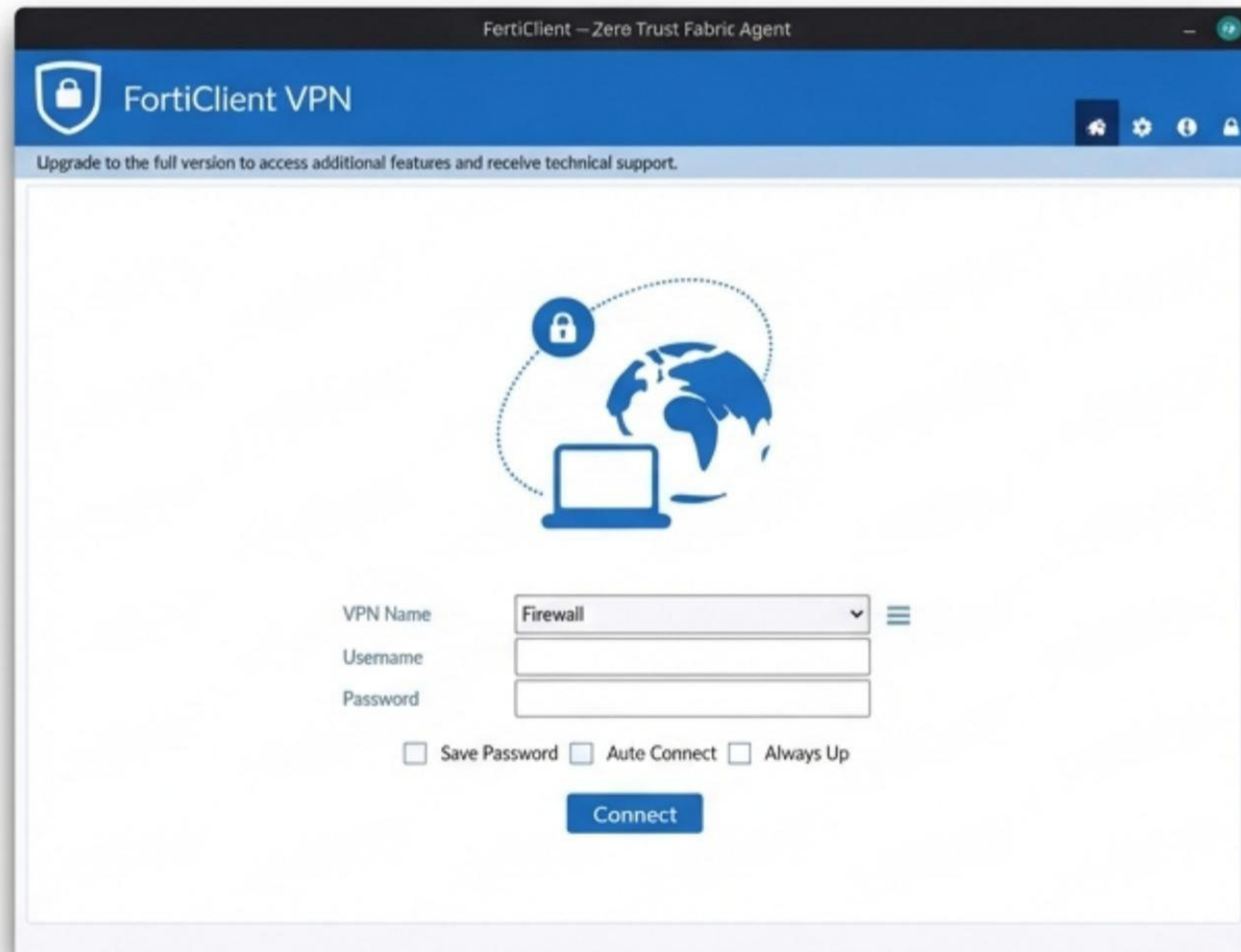


Threat Prevention

IPS Signatures,
Antivirus
Inspection &
DNS Sinkholing

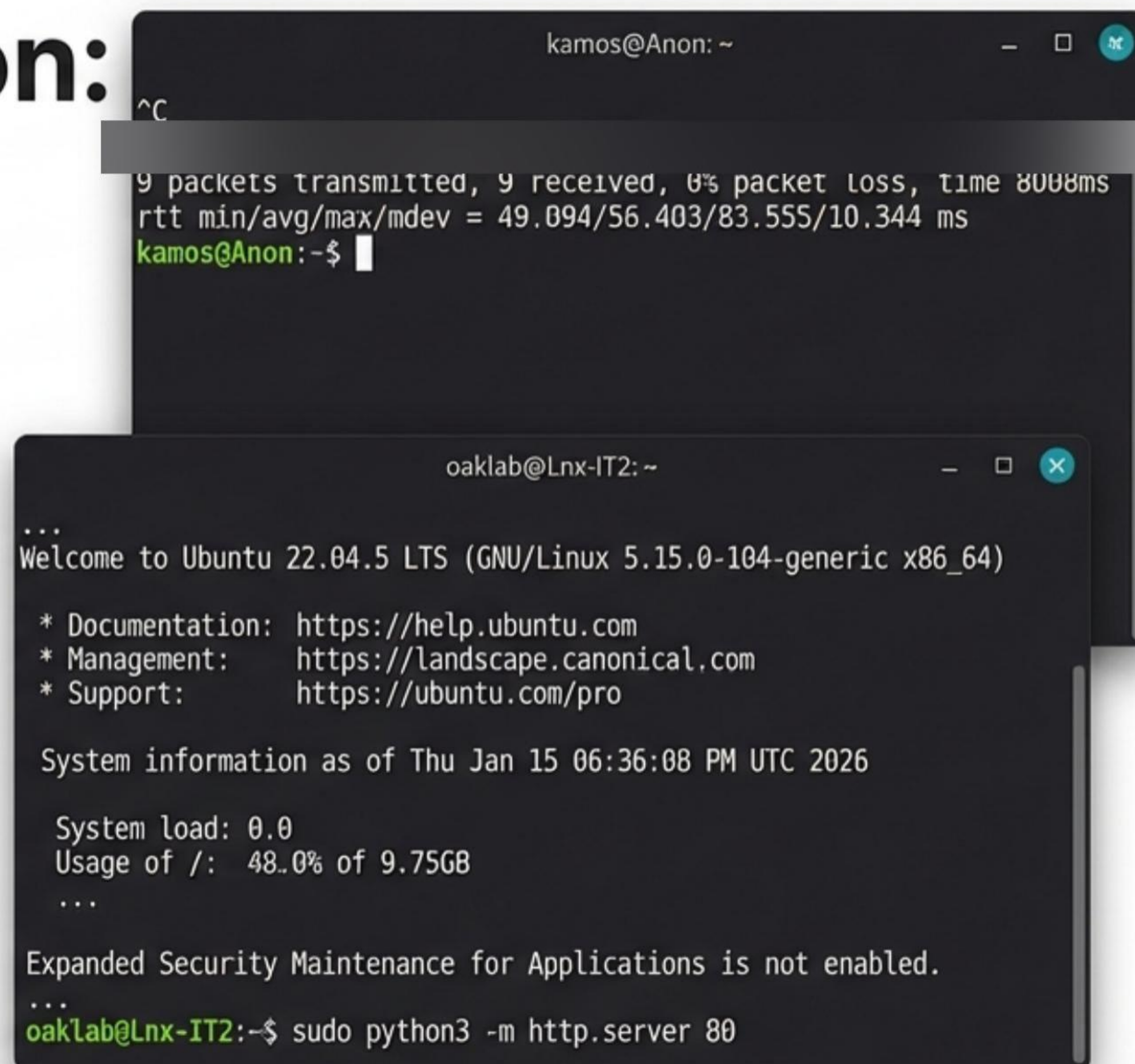
Perimeter Security: SSL-VPN Configuration

Remote access is secured via FortiClient VPN, establishing an encrypted tunnel into the network fabric.



Internal Segmentation: Linux Environment.

Adhering to Zero-Trust principles, lateral movement is restricted is restricted. Policy explicitly permits only ICMP (ping) for diagnostics and SSH (Port 22) for management.



The image shows two overlapping terminal windows. The top window, titled 'kamos@Anon: ~', displays the output of a network test: '9 packets transmitted, 9 received, 0% packet loss, time 8008ms' and 'rtt min/avg/max/mdev = 49.094/56.403/83.555/10.344 ms'. The bottom window, titled 'oaklab@Lnx-IT2: ~', shows the Ubuntu 22.04.5 LTS login screen with system information, including the date 'Thu Jan 15 06:36:08 PM UTC 2026', system load, and disk usage. It also shows the command 'sudo python3 -m http.server 80' being executed.

```
kamos@Anon: ~  
^C  
9 packets transmitted, 9 received, 0% packet loss, time 8008ms  
rtt min/avg/max/mdev = 49.094/56.403/83.555/10.344 ms  
kamos@Anon:~$  
  
oaklab@Lnx-IT2: ~  
...  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-104-generic x86_64)  
  
* Documentation: https://help.ubuntu.com  
* Management:   https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Thu Jan 15 06:36:08 PM UTC 2026  
  
System load: 0.0  
Usage of /: 48.0% of 9.75GB  
...  
  
Expanded Security Maintenance for Applications is not enabled.  
...  
oaklab@Lnx-IT2:~$ sudo python3 -m http.server 80
```

Verified connectivity to Ubuntu 22.04 LTS host.

Protocol Isolation: Windows RDP Control

Windows environments are isolated with strict access controls. Remote Desktop Protocol (TCP/3389) is governed by a dedicated firewall service object.

Create New Service

Name

FiratCan Bekar = RDP

Comments

FiratCan Bekar = RDP20/255

Color

Change

Category

Remote Access

Protocol Options

Protocol Type

TCP/UDP/SCTP

Address

IP Range

FQDN

0.0.0.0

Destination Port

TCP

3389

-

High

Create New Policy

Name

FiratCanBekar-SALES Win-> IT Win

Incoming interface

Sales (port3)

Outgoing interface

IT (port5)

Source

SALES_WINDOWShet

(port5)

+

Destination

IT_WINDOWS

+

Schedule

always

Service

FiratCan Bekar = RDP

+

Action

ACCEPT

DENY

Inspection mode

Flow-based

Proxy-based

Inbound Traffic: Virtual IP (VIP) & Port Forwarding.

External access to internal web resources is managed via Static NAT.

New Virtual IP

Name: FiratCan Bekar = VIP_SALES_TO_IT_LIN

Comments: FiratCan Bekar = VIP_SALES_TO_IT_LINUX_8080 43/255

Color: Change

Network

Interface: Sales (port3)

Type: Static NAT FQDN

External IP address/range: 10.10.0.0/24

Map to IPv4 address/range: 10.10.0.0/24

☐ Optional Filters

☒ Port Forwarding

Protocol: TCP UDP SCTP ICMP

Port Mapping Type: One to one Many to many

External service port: 8080

Map to IPv4 port: 8080

OK Cancel

Service Verification: Python HTTP Server

To validate the VIP, a Python http.server was deployed. External requests were successfully forwarded to the internal application.

```
oaklab@lnx-IT2:~$ sudo python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [15/Jan/2026 20:45:58] "GET / HTTP/1.1" 200 -
```

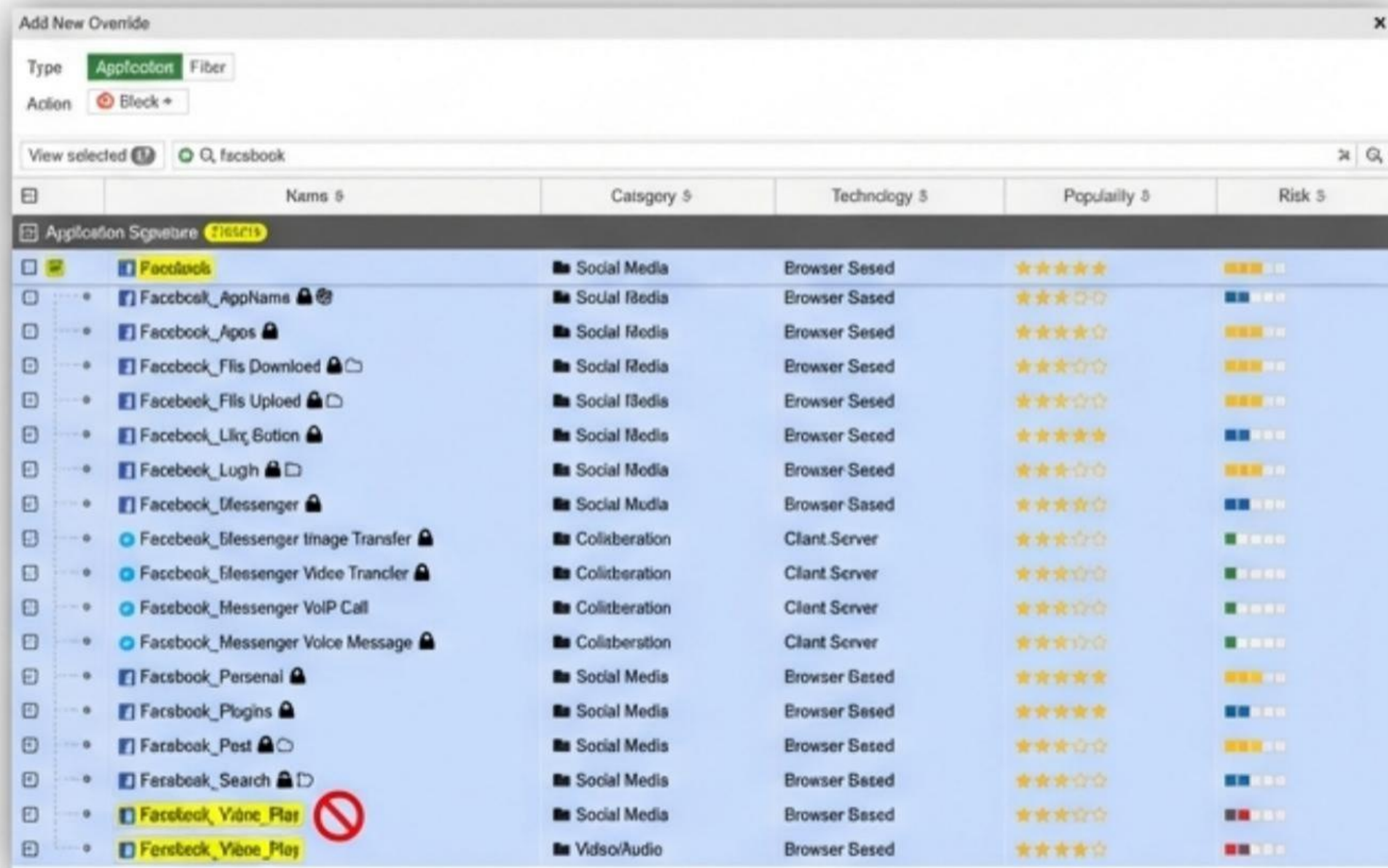
Traffic Flow

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
    <title>Directory listing for /</title>
  </head>
  <body>
    <h1>Directory listing for /</h1>
    <hr>
    <ul>
      <li><a href=".bash_history">.bash_history</a></li>
      <li><a href=".bash_logout">.bash_logout</a></li>
      <li><a href=".bashrc">.bashrc</a></li>
      <li><a href=".cache/">.cache/</a></li>
      <li><a href=".profile">.profile</a></li>
      <li><a href=".ssh/">.ssh/</a></li>
      <li><a href=".sudo_as_admin_successful">.sudo_as_admin_successful</a></li>
    </ul>
    <hr>
  </body>
</html>
```

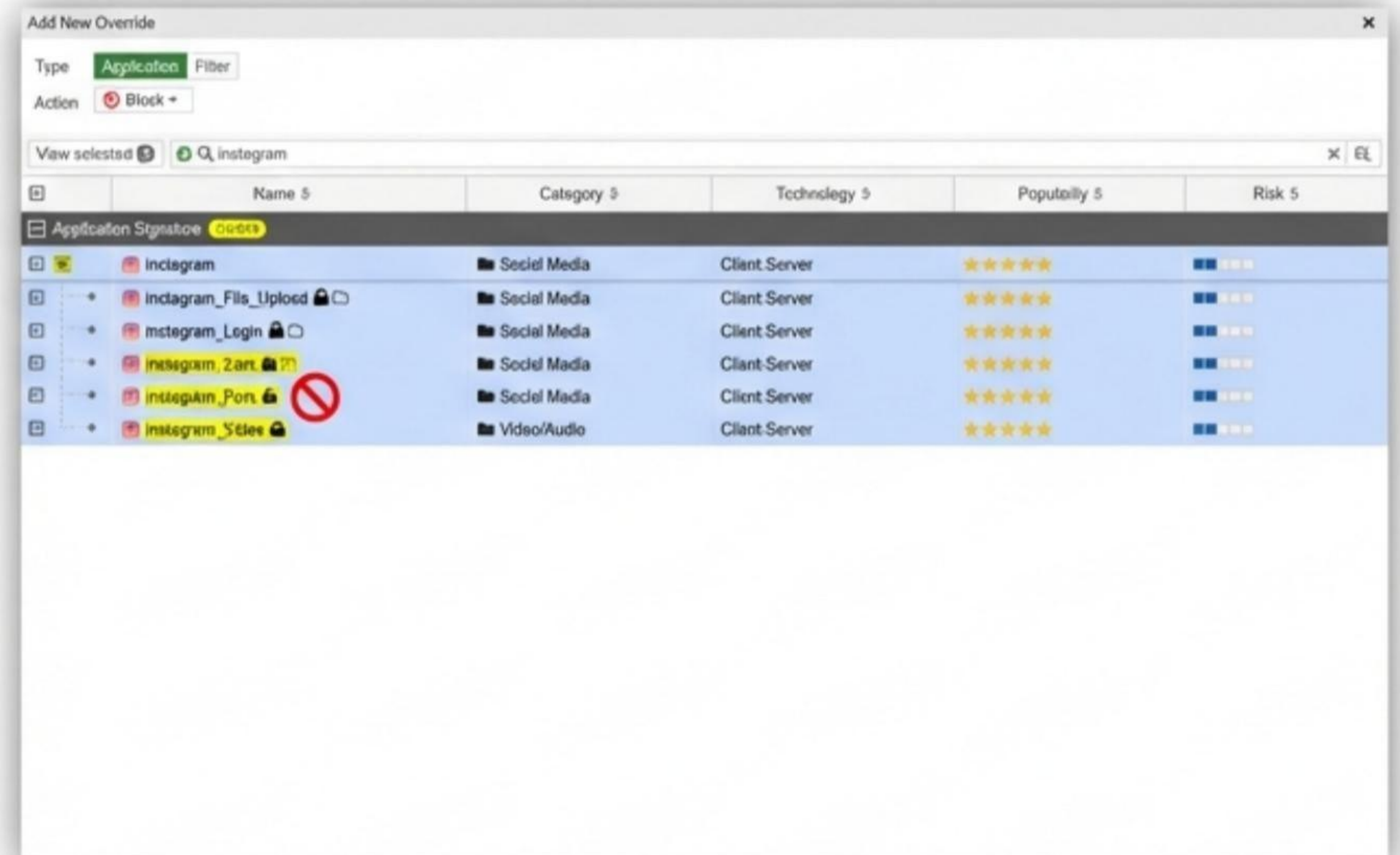
```
kamoe@Anon:~$
```


Layer 7 Application Control: Social Media.

Deep Packet Inspection (DPI) identifies traffic at the application layer. A rigid block policy is enforced for productivity-draining applications.



Name	Category	Technology	Popularity	Risk
Facebook	Social Media	Browser Based	★★★★★	★★★★★
Facebook_AppName	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Apos	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Files Download	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Files Upload	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Like Button	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Login	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Messenger	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Messenger Image Transfer	Collaboration	Client Server	★★★★★	★★★★★
Facebook_Messenger Video Transfer	Collaboration	Client Server	★★★★★	★★★★★
Facebook_Messenger VoIP Call	Collaboration	Client Server	★★★★★	★★★★★
Facebook_Messenger Voice Message	Collaboration	Client Server	★★★★★	★★★★★
Facebook_Personal	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Plugins	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Post	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Search	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Video_Play	Social Media	Browser Based	★★★★★	★★★★★
Facebook_Video_Play	Video/Audio	Browser Based	★★★★★	★★★★★

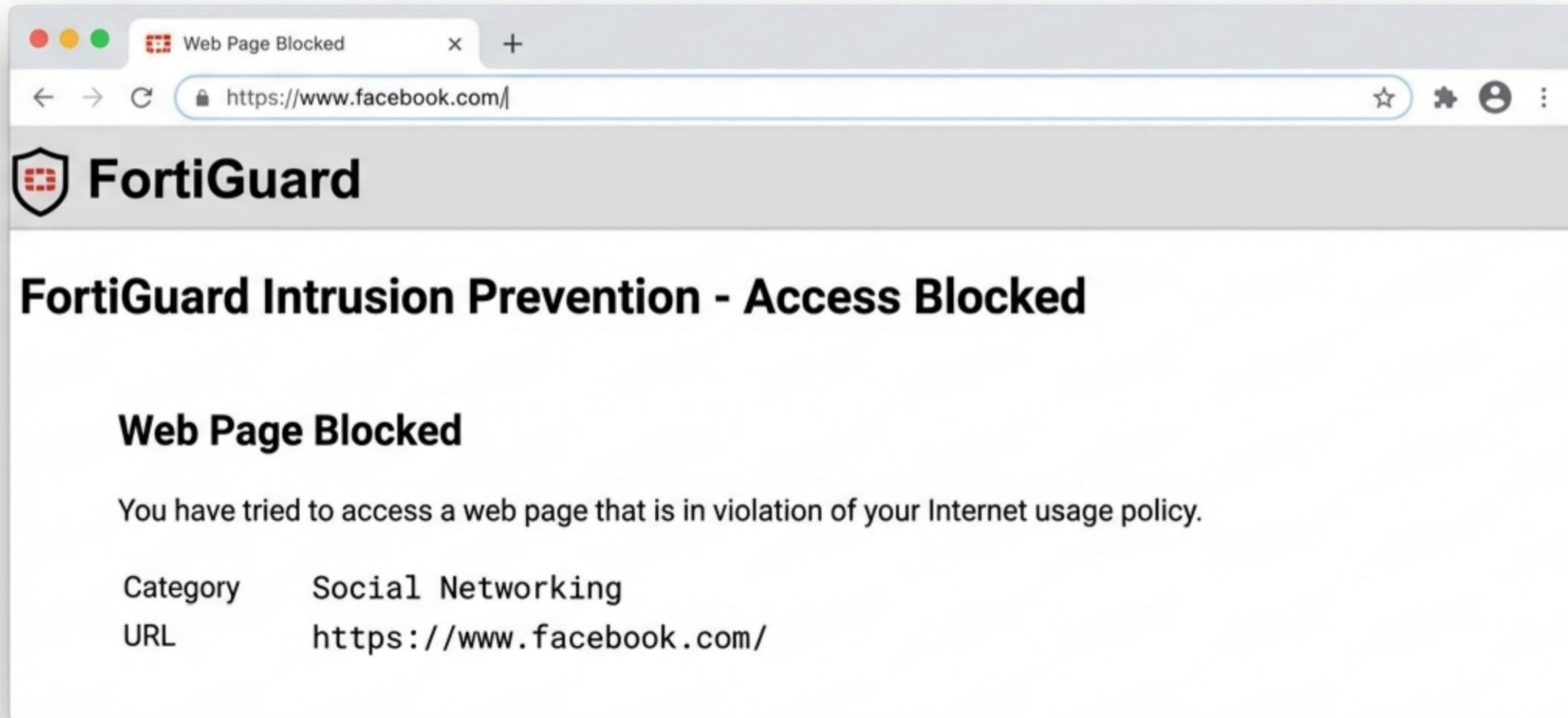


Name	Category	Technology	Popularity	Risk
Instagram	Social Media	Client Server	★★★★★	★★★★★
Instagram_Files Upload	Social Media	Client Server	★★★★★	★★★★★
Instagram_Login	Social Media	Client Server	★★★★★	★★★★★
Instagram_Post	Social Media	Client Server	★★★★★	★★★★★
Instagram_Post	Social Media	Client Server	★★★★★	★★★★★
Instagram_Stories	Video/Audio	Client Server	★★★★★	★★★★★

Signatures cover file uploads, login attempts, and video playback.

Policy Enforcement: User Experience

When a user attempts to access restricted social platforms, the connection is terminated immediately with a clear violation notice.



Corporate Web Filtering

Comprehensive URL filtering sanitizes outbound traffic to reduce liability and optimize network throughput.

FortiGuard Category Based Filter

Allow

Monitor

Block

Warning

Authenticate

Name	Action
Other Adult Materials	Block
Gambling	Block
Pornography	Block
Sports Hunting and War Games	Block
Bandwidth Consuming 1/5	
Streaming Media and Download	Block
General Interest - Personal 2/35	
Games	Block

Web Filter Violation

www.bet365.com



FortiGuard Intrusion Prevention - Access Blocked

Web Page Blocked

You have tried to access a web page that is in violation of your Internet usage policy.

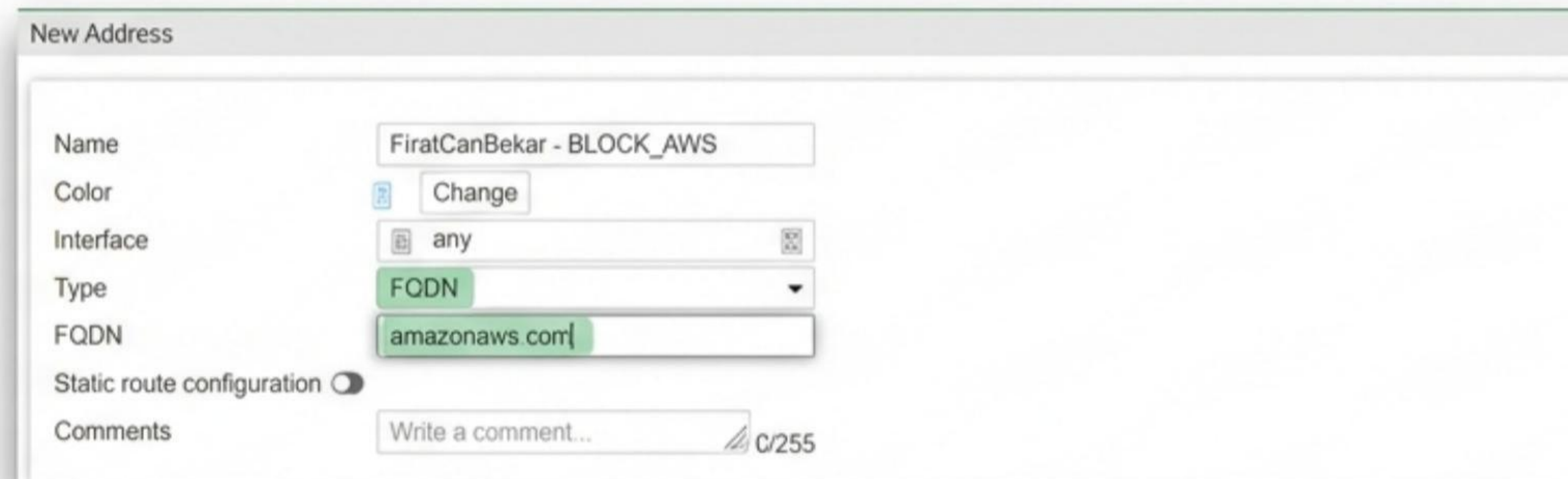
Category **Gambling**

URL **https://www.bet365.com/**

To have the rating of this web page re-evaluated [please click here](#)

Infrastructure Blocking: AWS Cloud Restrictions.

Traffic to specific cloud infrastructure is restricted using FQDN address objects to prevent unauthorized data exfiltration.



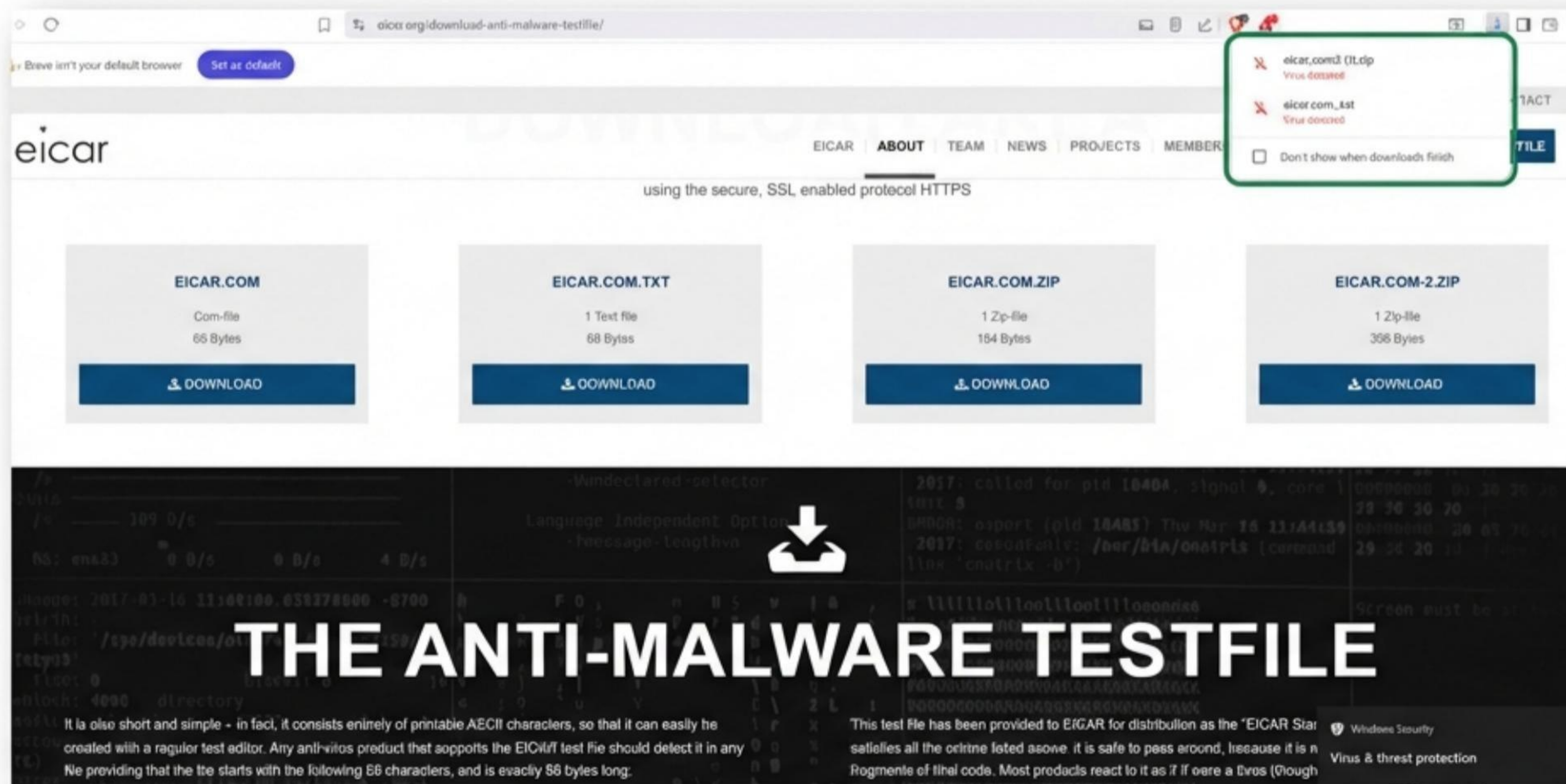
New Address

Name	FiratCanBekar - BLOCK_AWS
Color	 Change
Interface	 any 
Type	FQDN
FQDN	amazonaws.com
Static route configuration	☐
Comments	Write a comment... 0/255

```
oaklab@Lnx-IT2: ~  
oaklab@Lnx-IT2:~$ curl https://aws.amazon.com  
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">  
<TITLE>301 Moved</TITLE></HEAD><BODY>  
<H1>301 Moved</H1>  
The document has moved  
<A HREF="http://www.google.com/">here</A>.  
</BODY></HTML>  
oaklab@Lnx-IT2:~$
```


Active Threat Protection: Antivirus.

The FortiGate Antivirus profile actively scans HTTP/FTP streams. Successful detection of the EICAR test file confirms payload inspection is active.



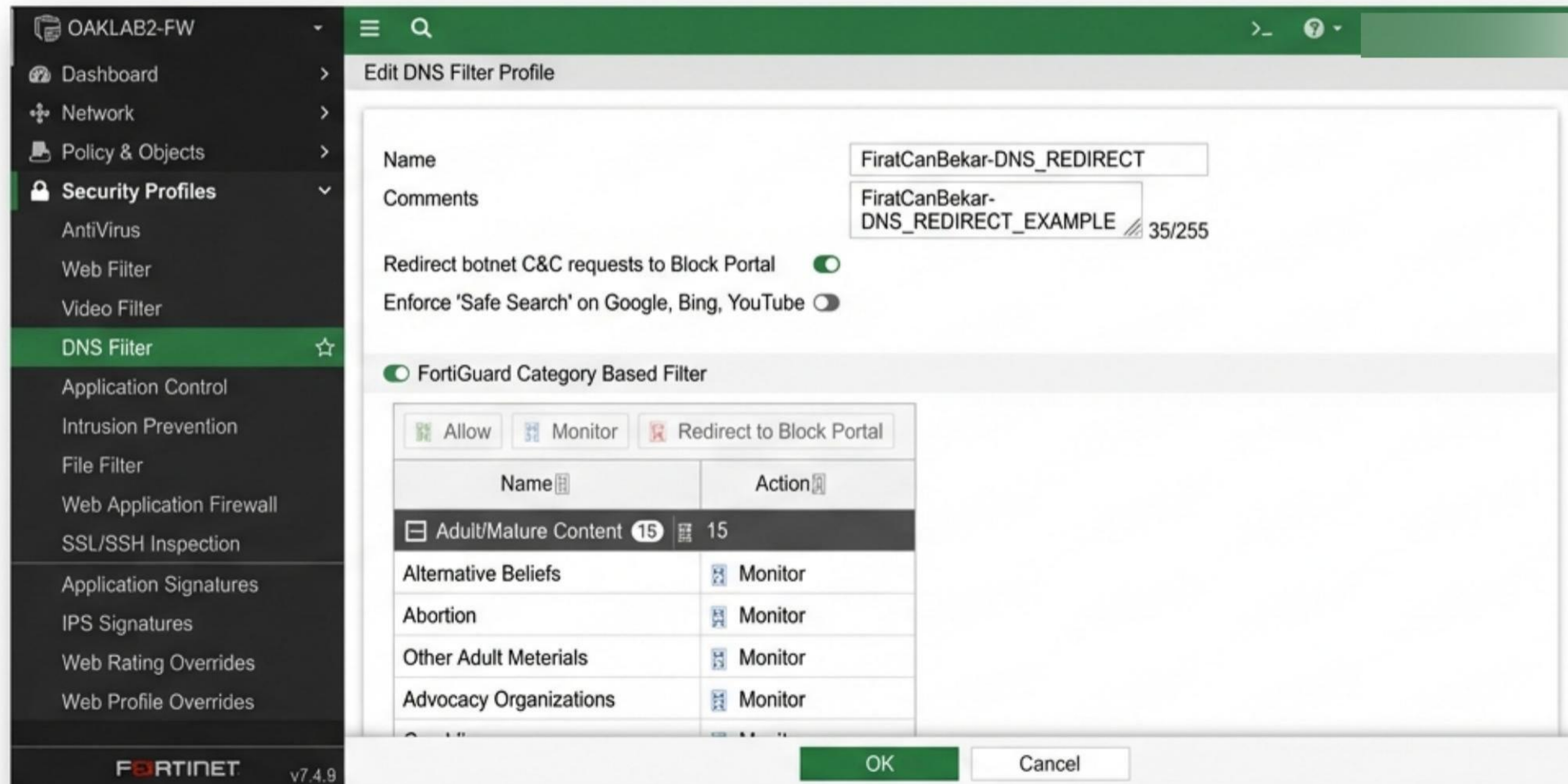
Intrusion Prevention System (IPS)

Network-based virtual patching is applied via IPS sensors to drop packets containing known exploit signatures.

IPS Signatures		Search	
Search			
☐	IPS Signature	IPS Signature	Action
☐	1Panel.image_repo.Create.Command.Injection 1Panel.image_repo.Create.Command.Injection	CVE-2023-36457	Block
☐	3Com.3CDaemon.FTP.Server.Buffer.Overflow 3Com.3CDaemon.FTP.Server.Buffer.Overflow	CVE-2005-0277	Block
☐	3Com.3CDaemon.FTP.Server.Buffer.Overflow 3Com.3CDaemon.FTP.Server.Buffer.Overflow	CVE-2005-0277	Block
☐	3Com.3CDaemon.FTP.Server.Buffer.Overflow 3Com.3CDaemon.FTP.Server.Buffer.Overflow	CVE-2005-0258	Block
☐	3Com.3CDaemon.FTP.Server.Buffer.Overflow	CVE-2005-0367	Block

DNS Security: Sinkhole Configuration.

To identify compromised internal hosts, the DNS filter intercepts requests to known botnet domains and redirects them to a monitored Sinkhole IP.



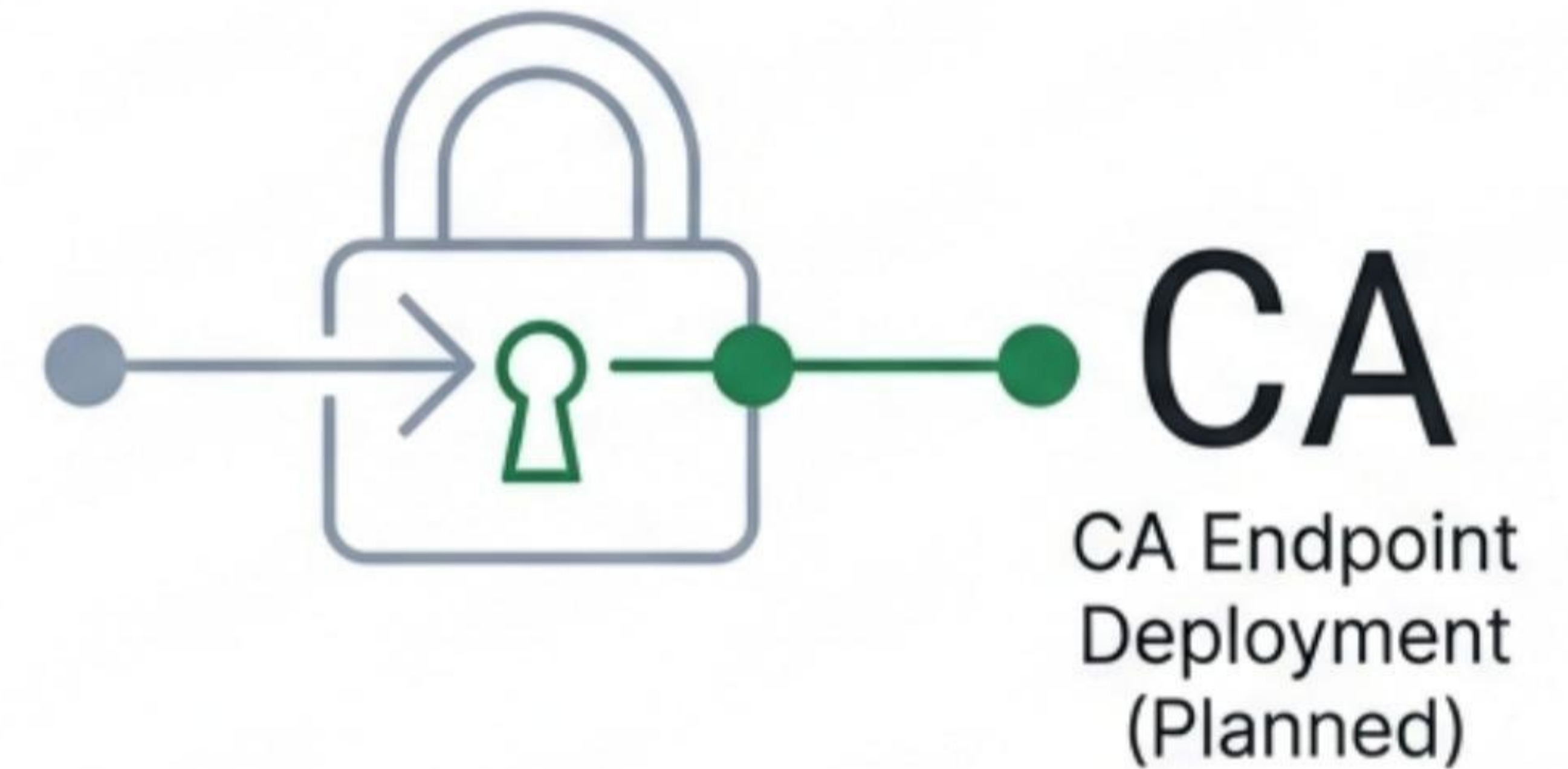
Visibility & Logging

Comprehensive logging is enabled on all firewall policies to ensure **auditability**. Every allowed connection and blocked violation is recorded.

Status	Service	Raw Content
(200 OK)	HTTP	HTTP/1.1 200 OK Accept-Ranges: bytes Content-Length: 783 Content-Type: text/html Date: Sat, 17 Jan 2026 11:28:25 GMT ETag: "b8197ad8a087dc1:0" ETag: "b8197ad8a087dc1:0" Last-Modified: Sat, 17 Jan 2026 11:03:01 GMT Serve...

Technical Constraints & SSL Inspection

In this lab environment, Deep SSL Inspection was **partially implemented**. While firewall logs successfully detected application signatures, full content blocking on HTTPS pages relies on Certificate Authority (CA) deployment on endpoints, which is a planned future enhancement.



Project Status: Completed.

- ✓ SSL-VPN Remote Access
- ✓ **Zero-Trust** Segmentation (Linux/Windows)
- ✓ Virtual IP & Port Forwarding
- ✓ Layer 7 Application Control
- ✓ Corporate Web Filtering
- ✓ AWS/Cloud FQDN Blocking
- ✓ Antivirus & IPS Deployment
- ✓ DNS Sinkhole Setup